

MULTI-SOURCE CANDIDATE DATA TRANSFORMER

TECHNICAL DESIGN SPECIFICATION

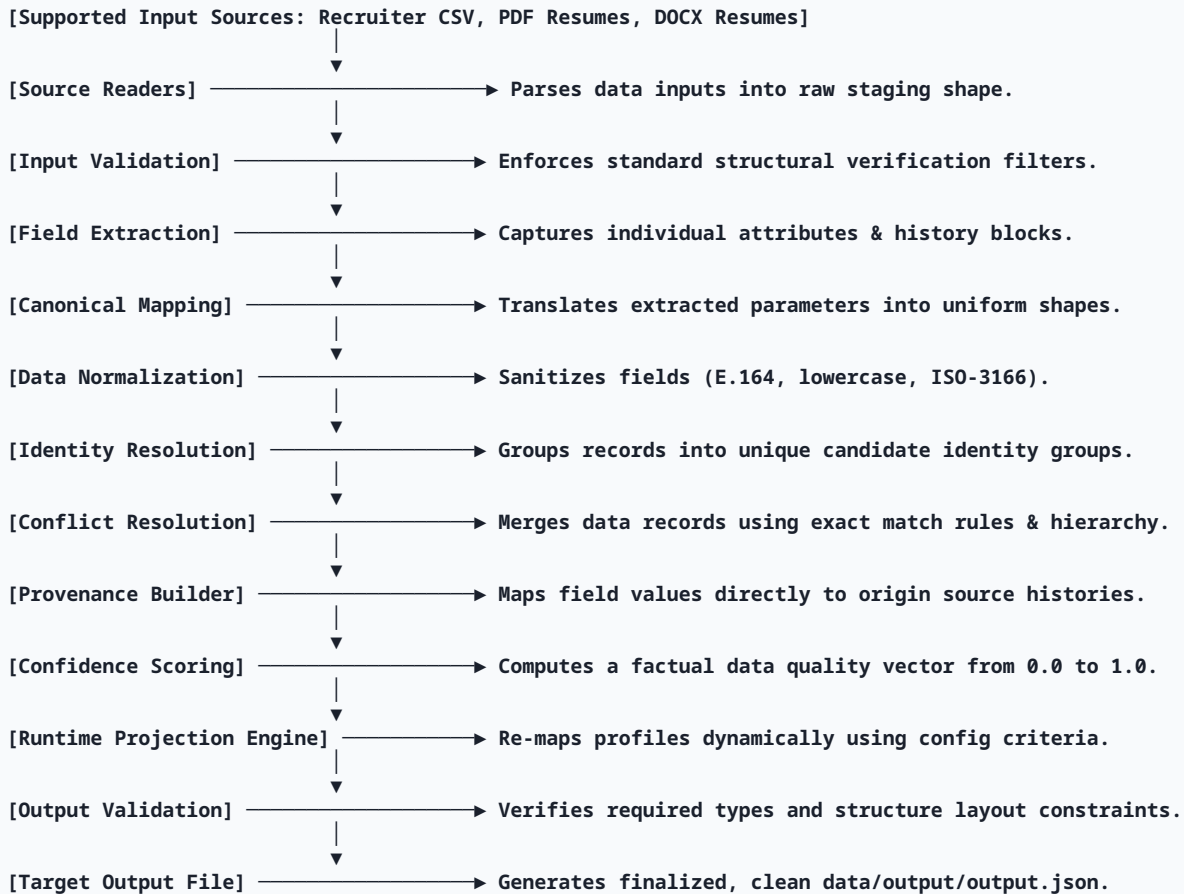
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- **GitHub Repository:** <https://github.com/UJJWAL2282/Multi-Source-Candidate-Data-Transformer.git>
- **Date:** July 2026

1. PROBLEM STATEMENT & SYSTEM OVERVIEW

- Recruitment platforms face major inefficiencies due to candidate files scattered across highly fragmented, heterogeneous sources.
- Data origins differ fundamentally in text structure, custom naming conventions, profile completeness, and input data quality.
- The same candidate frequently appears multiple times across various vectors with inconsistent, stale, or conflicting information blocks.
- The purpose of this architecture is to deliver a fully deterministic, explainable, and runtime-configurable data pipeline.
- The pipeline unifies multi-source data parameters into a single standardized golden representation called the Master Candidate Record (MCR).
- The engine integrates data streams, standardizes values, tracks data lineage, and assesses profile validity indicators.
- Core processing layers isolate canonical data rules completely from external output layouts to maximize backend pipeline stability.

2. PIPELINE WORKFLOW ARCHITECTURE

- The structural conversion pipeline processes inputs sequentially through precise specialized micro-service layers:



3. TECHNOLOGY INFRASTRUCTURE STACK

- **Python 3.11+:** Deployed as the primary execution environment and baseline interpreter framework runtime.
- **Pydantic:** Deployed for strict structural schema enforcement, semantic data type validations, and model validation checks.
- **pdfplumber:** Integrated for granular character mapping, coordinate text parsing, and character mining from raw PDF documents.
- **python-docx:** Parsing agent utilized for mining structural paragraph and xml text node clusters inside native DOCX documents.
- **spaCy:** Advanced natural language engine mapping entity segmentation tasks using the lightweight en_core_web_sm pipeline.
- **dateparser:** Deterministic date string normalization utility converting highly variable text structures into standard records.
- **pytest:** Automated validation framework running technical suite tests and structural regression execution verifications.

4. WORKSPACE STRUCTURE & SYSTEM INSTALLATION

```
multi-source-candidate-data-transformer/  
├── config/           # Configuration layers (pipeline_config.json, projection_config.json)  
├── data/             # Data stream containers (input/recruiter.csv, input/resumes/, output/)  
├── models/           # Pydantic core data types and canonical models  
├── pipeline/         # Sequential transformation workflow components  
└── tests/            # Quality validation scripts and framework regression test suites
```

```
# Repository deployment and environment initialization instructions  
git clone https://github.com/UJJWAL2282/Multi-Source-Candidate-Data-Transformer.git  
cd "CANDIDATE-TRANSFORMATION-PIPELINE"  
python -m venv .venv  
source .venv/bin/activate # On Windows terminals use: .venv\Scripts\activate  
pip install -r requirements.txt  
python -m spacy download en_core_web_sm  
python main.py --csv data/input/recruiter.csv --resumes data/input/resumes --output data/output/output.json  
pytest -v
```

5. CANONICAL OUTPUT SCHEMA & NORMALIZATION STRATEGY

FIELD ELEMENT	STAGING TYPE	NORMALIZATION & DATA RULE SPECIFICATION
candidate_id	String (UUID)	Unique pipeline-assigned identifier tracking individual candidate blocks.
full_name	String	Trim whitespace, strip duplicate inner spaces, standardize title case.
emails[]	Array of Strings	Converted to absolute lowercase format; verified using validation syntax.
phones[]	Array of Strings	Extraneous symbols and characters removed; formatted to strict E.164 rules.
location	Nested Object	Tracks structural sub-fields for City, Region, and ISO 3166 Alpha-2 entries.
skills[]	Array of Strings	Variant synonymous strings aligned onto canonical lookups via lookup dictionaries.
experience[]	Array of Objects	Structured employment blocks mapping timeline parameters, role titles, and dates.
provenance[]	Array of Objects	Auditable ledger components tying specific output elements to exact origin lines.
overall_confidence	Float Fractional	Factual profile completeness measurement clamped explicitly inside range [0.0, 1.0].

6. MERGE POLICY & CONFLICT RESOLUTION

- **Deterministic Priority Match Chain:** Groups candidate entities step-by-step using precise logical checks in sequence: 1. Exact email → 2. Exact phone → 3. Name + email → 4. Name + phone.
- **Single-Value Winner Selection Rules:** Resolves property value collisions by systematically prioritizing structured Recruiter CSV values over unstructured resume text variables.
- **Inter-Resume Tie-Breaking Policy:** When sorting conflicts across conflicting resume records, the engine preserves the most complete object block.
- **Fallback Precedence Order:** Resolves remaining deadlocks by choosing the most recent data metric update, then defaulting to the longest, most specific parameter string.
- **Pre-Merge Source Deduplication:** Identical within-source duplicate properties are completely isolated and filtered out using exact matching keys before triggering merge routines.

7. CONFIDENCE SCORING FRAMEWORK & PRESENTATION PROJECTION

- **Observable Metric Evaluation:** Computes profile data correctness metrics across a linear range [0.0, 1.0] using functional facts rather than opaque statistical weights.
- **Mathematical Base Metric Initialization:** Starts pipeline calculation runs from a fixed foundational baseline score defined as S_base = 0.50.
- **Incremental Core Modifiers:** Adds compounding increments: +0.15 for valid email, +0.10 for valid phone, +0.10 for name present, and +0.05 for current title present.
- **Multi-Source Agreement Bonus:** Appends a +0.05 score modifier for clear multi-source cross-alignment patterns or compiled skill profiles matching text count ≥ 3.
- **Critical Property Deductions:** Subtracts a strict penalty reduction of -0.10 for each missing essential profile category block, appending the final clamped product as overall_confidence.
- **Dynamic External Presentation Projection:** Utilizes projection_config.json configurations to decouple core database models from changing target client shapes at runtime.
- **Dynamic Output Configuration Controls:** Configures dynamic key filtering transformations, custom label renames, property flattening tiers, metadata visibility tags, and missing value default settings.
- **Post-Projection Compliance Assertions:** Validates essential keys, verifies data types, and strictly rejects unexpected parameters before writing final files to data/output/output.json.

8. EDGE CASES & OPERATIONAL SCOPE

- **Missing Email Fallback Routing:** Routes identity groupings through phone channel chains or name+phone parameter blocks when a valid email address is absent.
- **Corrupt Input Isolation:** Logs detailed extraction errors and skips unreadable files safely without causing batch process failures.
- **Incomplete Recruiter Vector Rows:** Discards corrupted indices or parses valid fields to capture partial information.
- **Same-Name Profile Isolation:** Evaluates distinct contact details to ensure same-name candidates with separate details remain isolated as unique profiles.
- **Out of Scope System Boundaries:** Excludes machine learning fuzzy matching heuristics, live continuous streaming updates, image-based unreadable resume layout OCR parsing, external live scraping sync connections (LinkedIn/GitHub sync APIs), database administration, and graphical review UI dashboards.

9. CONCLUSION

- The Multi-Source Candidate Data Transformer delivers a powerful, production-ready backend engine for consolidating highly fragmented talent profiles into an explainable, unified single source of truth.
- By leveraging strict deterministic matching, canonical-first design principles, and an externalized configuration layer, the architecture completely isolates standard consolidation business logic from evolving presentation layer constraints.
- The system handles edge cases like file corruption or partial data vector entries gracefully, optimizing data completeness and system robustness without sacrificing performance.
- The factual confidence evaluation matrices and multi-layered provenance tracking tables maximize visibility and auditability, making the pipeline highly transparent and maintainable for institutional scale deployments.